01 EXECUTIVE ARCHITECTURE OVERVIEW

NasonNation // Command is a self-hosted, zero-cloud computer vision, facial biometric, and Automatic License Plate Recognition (ALPR) platform running on bare-metal Debian Linux (192.168.1.x). Live RTSP camera streams are ingested once by MediaMTX and fanned out over sub-second WebRTC (WHEP) to web clients and an asynchronous Python FastAPI edge CV engine (OpenCV MOG2, YOLOv8s, and YuNet + SFace 128-D embeddings). Heavy multimodal scene reasoning and plate OCR are offloaded over gigabit LAN to a dedicated NVIDIA RTX 3060 Ti node running Qwen2.5-VL via Ollama.

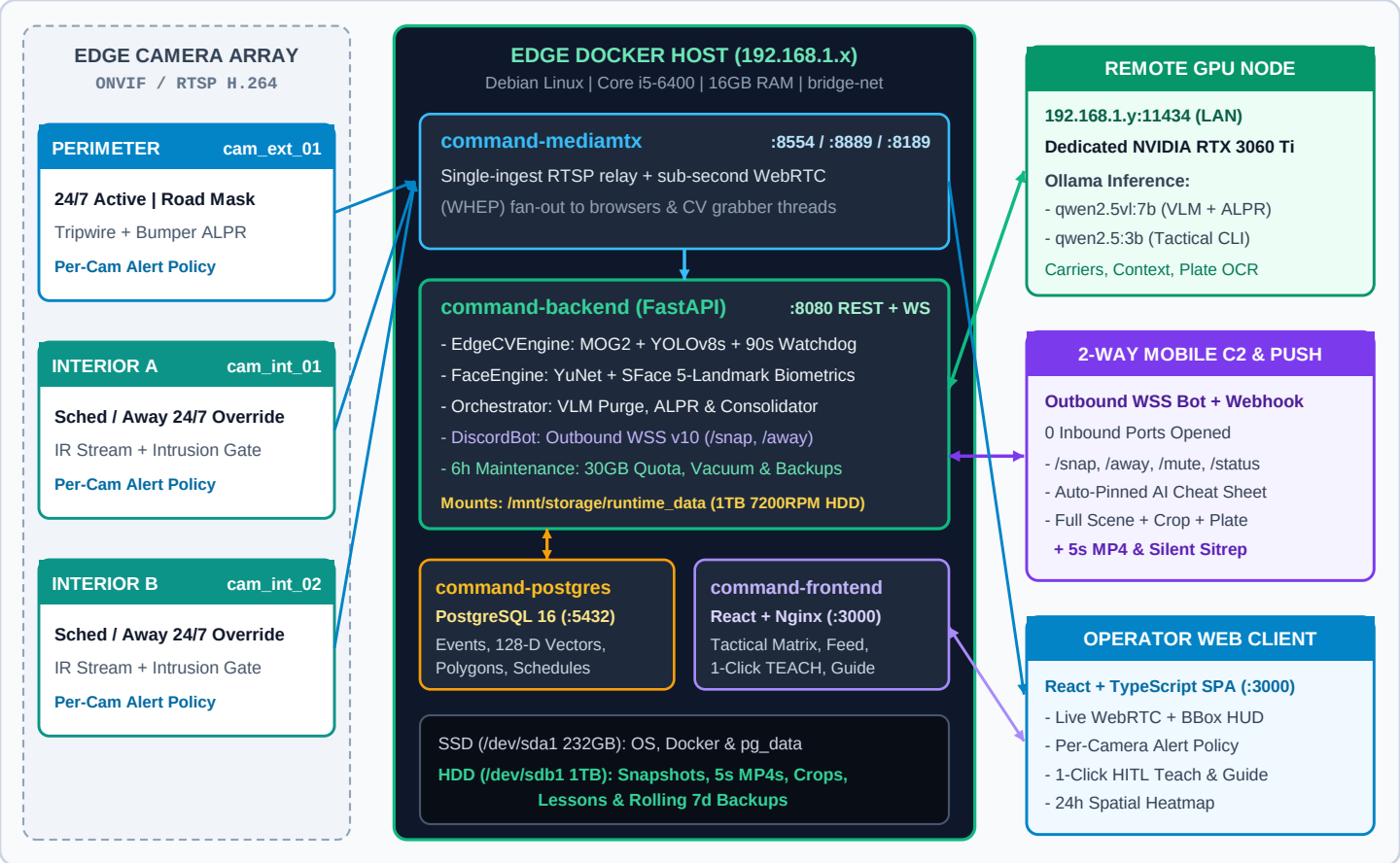
MULTI-STREAM RTSP
24/7 + 1-Click Away Mode

SPLIT-NODE AI
Edge YOLOv8s + RTX 3060 Ti

128-D BIOMETRICS
5-Landmark Gate + RBAC

ZERO-PORT 2-WAY C2
Outbound Bot + MP4 Push

02 DISTRIBUTED SYSTEM & NETWORK TOPOLOGY



03 CORE ARCHITECTURAL PRINCIPLES

Split-Compute Efficiency: Lightweight MOG2 motion gating and YOLOv8s ONNX run on the edge CPU, waking the remote RTX 3060 Ti VLM only when verified physical displacement occurs outside road exclusion masks.

Shared Global Biometrics: A single 128-D YuNet + SFace vector gallery in PostgreSQL/RAM serves all camera streams concurrently. Enrolling a trusted user once immediately identifies them across all exterior and interior zones.

Hardware Wear Isolation: High-churn MP4 video clips, high-res snapshots, face crops, and daily compressed backups bind-mount to a 1TB 7200RPM HDD (/dev/sdb1), protecting the 232GB SATA boot SSD (/dev/sda1) from write wear.

04 6-STAGE REAL-TIME DETECTION, BIOMETRIC & VLM PIPELINE

STAGE 1 **RTSP Ring-Buffer & MOG2 Motion Gate**

LatestFrameGrabber maintains a rolling 48-frame (960x540) RAM ring buffer per stream for instant 5-second MP4 pre/post-event replays. OpenCV MOG2 filters static scenes; polygon Exclusion Masks suppress street traffic before neural inference runs.

STAGE 2 **YOLOv8s Detection, True Displacement & Parked-Vehicle Lock**

ONNX Runtime executes YOLOv8s for person, vehicle, and animal classes. ParkedVehicleRegistry locks stationary driveway vehicles after 6 sightings (IoU >= 0.68). Spatial co-occurrence heuristics distinguish walked dogs from solo neighborhood cats.

STAGE 3 **Hardened 5-Landmark YuNet + SFace Biometrics & Guardrails**

Enforces upper-body ROI + 5-landmark anatomical geometry & Laplacian sharpness before matching 128-D SFace embeddings against Whitelist roles, Visitors, and public Watchlist feeds. Role-based guardrails lock external OSINT tools on trusted identities.

STAGE 4 **Remote RTX 3060 Ti VLM Inspection, False-Alarm Purge & ALPR**

deep_inspect_detection() sends target + bumper crops to Qwen2.5-VL over LAN (with 30s edge failover cache). Verifies face visibility, purges inanimate false triggers, extracts [PLATE: ...] OCR, and applies Carrier/Service rules (Alert vs Silent).

STAGE 5 **1-Click Human-in-the-Loop Active Learning (TEACH & ID FACE)**

Operator UI & Field Guide provide 1-click corrective teaching. Teaching STRANGER purges corrupted dynamic embeddings and restores breach flags; confirming a trusted user auto-enrolls clear portraits and injects camera-specific VLM prompt lessons.

STAGE 6 **Track-Scoped Encounter Consolidator & 2-Way Discord C2**

Selects the single highest-confidence frame across +-30s bursts (with per-track owner shielding so concurrent threats never hide), pushing scene + crop + plate + 5s MP4. Outbound Gateway v10 Bot powers /snap, /away, /mute, /status & auto-synced Cheat Sheet.

05 ROLE-BASED ALERT ROUTING & PER-CAMERA POLICY MATRIX

SUBJECT ROLE / RULE CLASS	MODE: WHITELIST-SILENT	MODE: ALL-ALERTS ACTIVE	MULTIPART PUSH PAYLOAD
PRIMARY_ADMIN (Same Track)	SILENT (Track-Scoped)	SILENT (Admin Exemption)	Never alerts (0 self-spam)
HOUSEHOLD_WHITELIST (A/B/C)	SILENT (Trusted Role)	ALERTS (Household Arrival)	Conf % + Crop + 5s MP4
Watchlist Biometric Match	URGENT ALERT	URGENT ALERT	Priority Ping + Crop + MP4
Perimeter / Indoor Breach	HIGH ALERT	HIGH ALERT	Priority Ping + Crop + MP4
Logistics Carrier (UPS/AMZN)	DELIVERY ALERT	DELIVERY ALERT	Carrier + Plate + 5s MP4
Scheduled Service Crew	SILENT (Visitor crop purged)	SILENT (Service Rule)	Indexed in Event DB only
Unclassified Vehicle / Animal	Indexed in Event DB only	ALERTS (Full-Audit Mode)	Conf % + Plate + 5s MP4

POSTURE & SCHEDULE ENGINE (HOME vs. 1-CLICK AWAY MODE):

Supports 24/7, Pause, Schedule, and 1-Click AWAY MODE (24/7 all-camera override + exit grace) without overwriting Home schedules.

06 DUAL-DISK STORAGE ARCHITECTURE (NAND SSD WEAR PROTECTION)

TIER 1: /dev/sda1 - 232 GB SATA SSD

Role: Low-Latency OS, Containers & SQL Indexes

- Debian Linux Root (/) & Systemd (fstrim.timer active)
- Docker Engine & Immutable Application Layers (:ro)
- pg_data Volume: PostgreSQL 16 tables & 128-D indexes
- ONNX Runtime & YOLOv8s / YuNet / SFace weights
- JSON Container Logs capped at 10MB x 3 files/service
- Isolated from 24/7 sequential video/image write wear

TIER 2: /dev/sdb1 - 1 TB 7200RPM HDD

Role: Sequential Media, Runtime State & DR Backups

- Dedicated high-endurance sequential storage mount
- runtime_data/snapshots -> /tmp/command_snapshots
  (crop_, full_, plate_ JPEGs + 5s H.264 .mp4 clips)
- runtime_data/faces -> /tmp/command_faces
  (enrolled face crops, delivery_rules & ai_lessons.json)
- runtime_data/backups -> Rolling 7-day .json.gz dumps

07 AUTOMATED SELF-HEALING MAINTENANCE (periodic_maintenance_worker)

1. 30 GB Media Quota & Stream Watchdog

Enforces a 30 GB ceiling / 14-day retention on Tier-2 HDD; 90s RTSP watchdog alerts on stream dropouts and posts silent daily Sitreps.

2. Gallery Sanitizer & Event Pruning

Startup sanitizer purges occluded/blurred dynamic embeddings; 6h worker prunes events and 1-sighting visitors >14 days old.

3. Automated VACUUM ANALYZE

Executes autocommit PostgreSQL VACUUM ANALYZE on high-churn tables every 6h to eliminate dead-tuple bloat and maintain query speed.

4. Cross-Drive Disaster Recovery

Writes daily gzipped JSON snapshots (embeddings, polygon zones, schedules, rules, and learned prompts) from Tier-1 SSD to Tier-2 HDD.

08 SERVICE INTERFACES & DEFENSE-IN-DEPTH SECURITY POSTURE

PORT / PROTO	SERVICE / NODE	ARCHITECTURAL PURPOSE & NETWORK BOUNDARY
`3000 / TCP`	**command-frontend**	Private LAN - React/TypeScript Tactical Matrix, Live HUD & Biometric Gallery
`8080 / TCP`	**command-backend**	Private LAN - FastAPI REST (/api/detections, /teach, /faces, /zones) + WS
`8889 / 8189`	**command-mediamtx**	Private LAN - WHEP HTTP Signaling (:8889) & Single-Port WebRTC ICE UDP (:8189)
`8554 / TCP`	**command-mediamtx**	Internal Bridge - RTSP camera ingest & low-overhead OpenCV frame grabbers
`5432 / TCP`	**command-postgres**	Internal Bridge - PostgreSQL 16 (command_db); cross-drive daily backups
`11434 / TCP`	**RTX 3060 Ti Node**	Private LAN - Dedicated Ollama GPU server for Qwen2.5-VL & license plate OCR

SECURITY, OPSEC & GOVERNANCE SUMMARY

1. Zero Inbound Attack Surface: Outbound-only Discord Gateway v10 Bot (guild-locked) + TLS webhooks; 0 router ports opened.
2. Secret & Biometric Isolation: Webhook URLs, Bot Tokens, RTSP credentials, DB volumes, and face crops stay strictly out of VCS.
3. Immutable Runtime & Privacy Locks: Source mounts read-only (:ro) in containers; external lookup tools auto-lock on trusted roles.